Name / Имя: _____ Class: _____ Date: _____

Which part of the body will be affected?

Textbook Activity p. 59. Work in pairs. Use the information box.

Information box · Информация

Element	What it does in the body (textbook pp. 58–59)
Calcium (Ca)	Found in bones and teeth. Acts in muscle contraction and blood clotting.
Phosphorus (P)	Found in bones and teeth and in nucleic acids. Helps in energy transmission.
Potassium (K)	Main role in the nervous system. Na, K and Cl form the nerve impulses.
Magnesium (Mg)	Role in protein production and relaxation of muscles.
Iron (Fe)	Part of hemoglobin — the blood protein that transports oxygen.
Iodine (I)	Main component of the thyroid hormones.
Fluorine (F)	A microelement. Teacher information: it helps keep teeth strong (not explained in the textbook).

Task A · Complete the table · Заполни таблицу

Word bank: **thyroid gland · blood · bones · teeth · muscles · nervous system**

Element	Macro or micro?	Part of the body affected	Because… (reason)
Iodine (I)			
Fluorine (F)			
Iron (Fe)			
Calcium (Ca)			
Phosphorus (P)			
Potassium (K)			

Example: If the body lacks iodine, the thyroid gland will be affected because iodine is the main part of thyroid hormones.

Task B · Read the chart (Figure 4.2.1) · Прочитай диаграмму

Element	O	C	H	N	Ca	P	K	S	Na	Cl	Mg
% of body	65	18.5	9.5	3.3	1.5	1	0.4	0.3	0.2	0.2	0.1

Microelements together make up less than 0.1%.

1. Which element is the most abundant in the body?

2. Add the percentages of O, C, H and N. Show your working. What do you notice?

3. Iron is less than 0.1% of the body. Is it still important? Explain.

Task C · Element detective · Детектив

A 12-year-old is growing fast. Which **two** elements help to build bones and teeth? Use the information box.

> I think _____ and _____ are needed because _____

Challenge: Nerve signals are weak. Which elements form nerve impulses?